STAT 315: Law of Large Numbers and Monte-Carlo

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Independent Identically Distributed (=IID) Random Variables

Operational Meaning: Repeat an experiment n times, each time under the exactly same conditions, each time independently of the other experiments. The outcome of each experiment is a measurement Y_i, $i = 1, 2, \cdots, n$ (a random variable).

IID RV and sample mean

- $Y1, \cdots, Y_n$ are n independent RV.
- $Y1, \cdots, Y_n$ are identically distributed: The PDF of each Y_i is $f(y_i)$ with the same f.
- The joint PDF is $f(y_1, \cdots, y_n) = f(y_1) \cdots f(y_n)$.
- The mean $E[Y_i] = \mu$ and the variance $V[Y_i] = \sigma^2$ do no depend of i.
- The sample mean is

$$\overline{Y} = \frac{Y_1 + \cdots + Y_n}{n}$$

The Law of Large Numbers

Remember

Mean and Variance

If $Y_1, \cdots Y_n$ IID with $E[Y_i] = \mu$ and $V[Y_i] = \sigma^2$ then

$$E[\overline{Y}] = E\left[\frac{Y_1 + \cdots Y_n}{n}\right] = \mu \qquad V[\overline{Y}] = V\left[\frac{Y_1 + \cdots Y_n}{n}\right] = \frac{\sigma^2}{n}$$

So by Chebyshev Inequality

Law of Large Numbers

If $Y_1, \cdots Y_n$ IID with $E[Y_1] = \mu$ and $V[Y_1] = \sigma^2$ then

$$P\left\{|\overline{Y} - \mu| > \epsilon\right\} \leq \frac{V[\overline{Y}]}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2} \underbrace{\rightarrow}_{\text{as } n \to \infty} 0$$

Interpretation of the Law of Large Numbers: Concentration of the sample mean $\overline{Y}$ around the true mean μ

The variance of $\overline{Y}$ decreases with n so the PDF of $\overline{Y}$ is more and more concentrated around the mean μ.

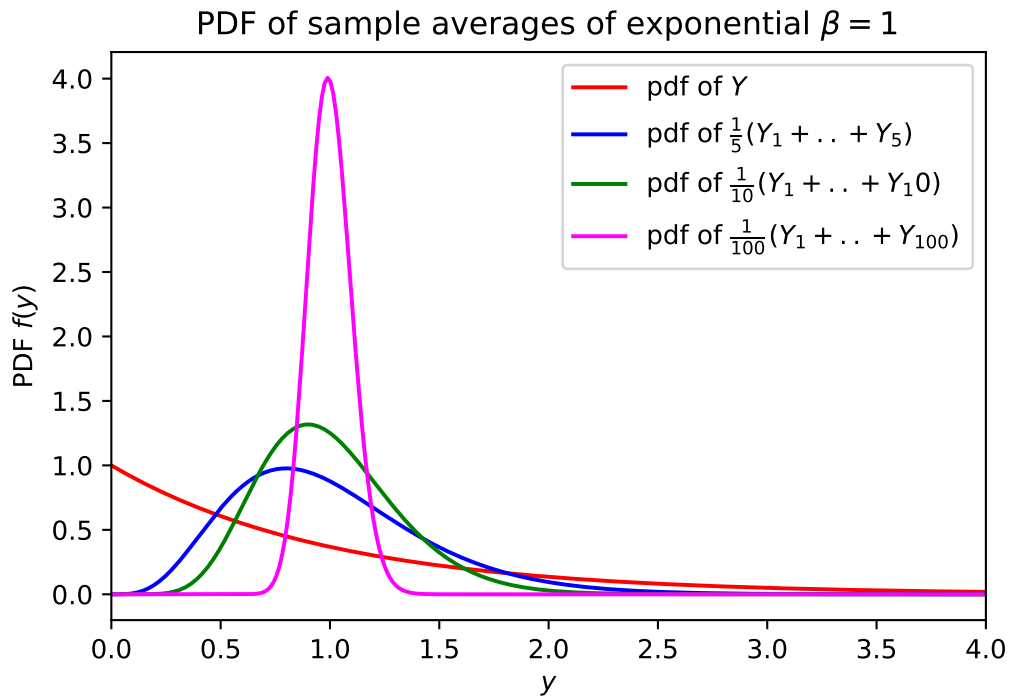


Figure: If Y_i is exponential with parameter 1 then $\frac{Y_1+\cdots+Y_n}{n}$ is gamma with $\alpha = 1$ and $\beta = \frac{1}{n}$ see previous slides.

Using the Law of Large numbers to estimate an unknown μ

Operational meaning: You can perform an experiment resulting a measurement Y but you do not know the PDF of Y neither do you know the mean μ? The Law of large numbers tells you to perform n independent

experiment $Y_1, Y_2, \cdots, Y_n$ and use the approximation

$$\frac{Y_1 + \cdots + Y_n}{n} \approx \mu$$

Chebyshev Theorem tells you what this is a good approximation with high probability as n grows.

$\longrightarrow$ Can we figure out how good this approximation really is? The central limit theorem provides one answer: all sample mean looks like normal random variables for large n.

The Monte-Carlo method

Monte-Carlo Method

Put the law of large numbers to work: Imagine μ is some quantity you try to evaluate.

- Find a random variable X with $E[X] = \mu$.
- Simulate $X_1, X_2, \cdots, X_n$ on a computer.
- Get the estimate $\mu \approx \frac{X_1 + \cdots + X_n}{n}$

Example: Estimate the number π using random numbers

- Let U and V be two random numbers. If $U^2 + V^2 \leq 1$ set $Y = 1$, otherwise set $Y = 0$. Then

$$P(Y = 1) = P(U^2 + V^2 \leq 1) = \frac{\pi}{4}$$

  and $E[Y] = \frac{\pi}{4}$.
- Taking IID copies $Y_1, X_2, , Y_n$ we have

$$4\frac{Y_1 + \cdots + Y_n}{n} \approx 4E[Y]$$

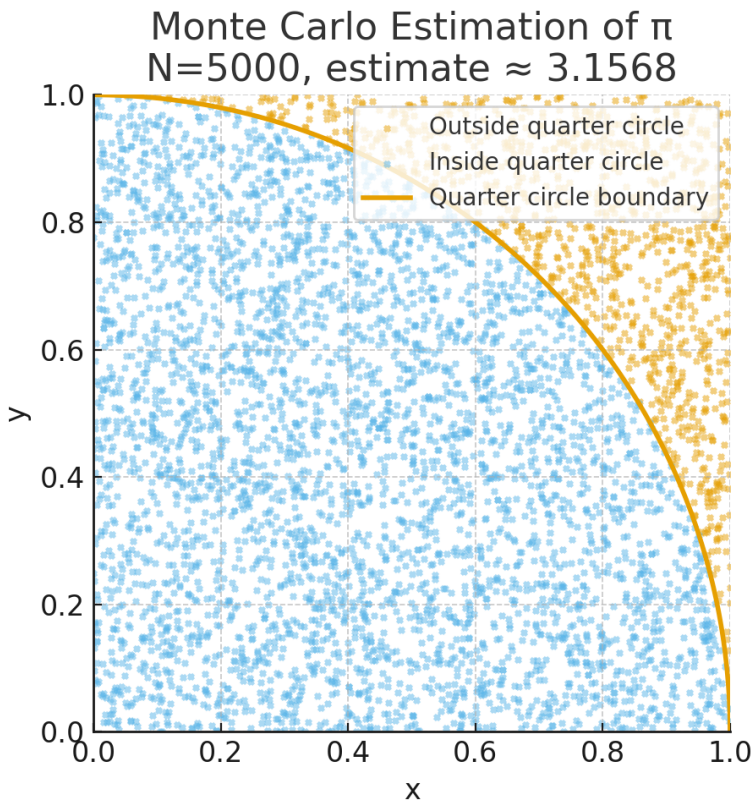
Monte Carlo Estimation of π

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

N = 5000
x = np.random.rand(N)
y = np.random.rand(N)

inside = x**2 + y**2 <= 1
pi_est = 4 * inside.mean()
```



Buffon's Needle Problem: Setup and Intuition

Buffon's Needle is one of the oldest problems in geometric probability (1733). It asks:

The Question

A needle of length ℓ is dropped at random on a floor with equally spaced parallel lines a distance d apart ($\ell \leq d$).

What is the probability that the needle crosses a line?

Randomness in the experiment:

- The *angle* Θ between the needle and the parallel lines (uniform on $[0, \pi/2]$).
- The *distance* X from the needle's center to the nearest line (uniform on $[0, d/2]$).

Needle crosses a line when the vertical projection exceeds X:

$$\text{Crossing occurs if } X \leq \frac{\ell}{2} \sin\Theta.$$

Geometry of Buffon's Needle

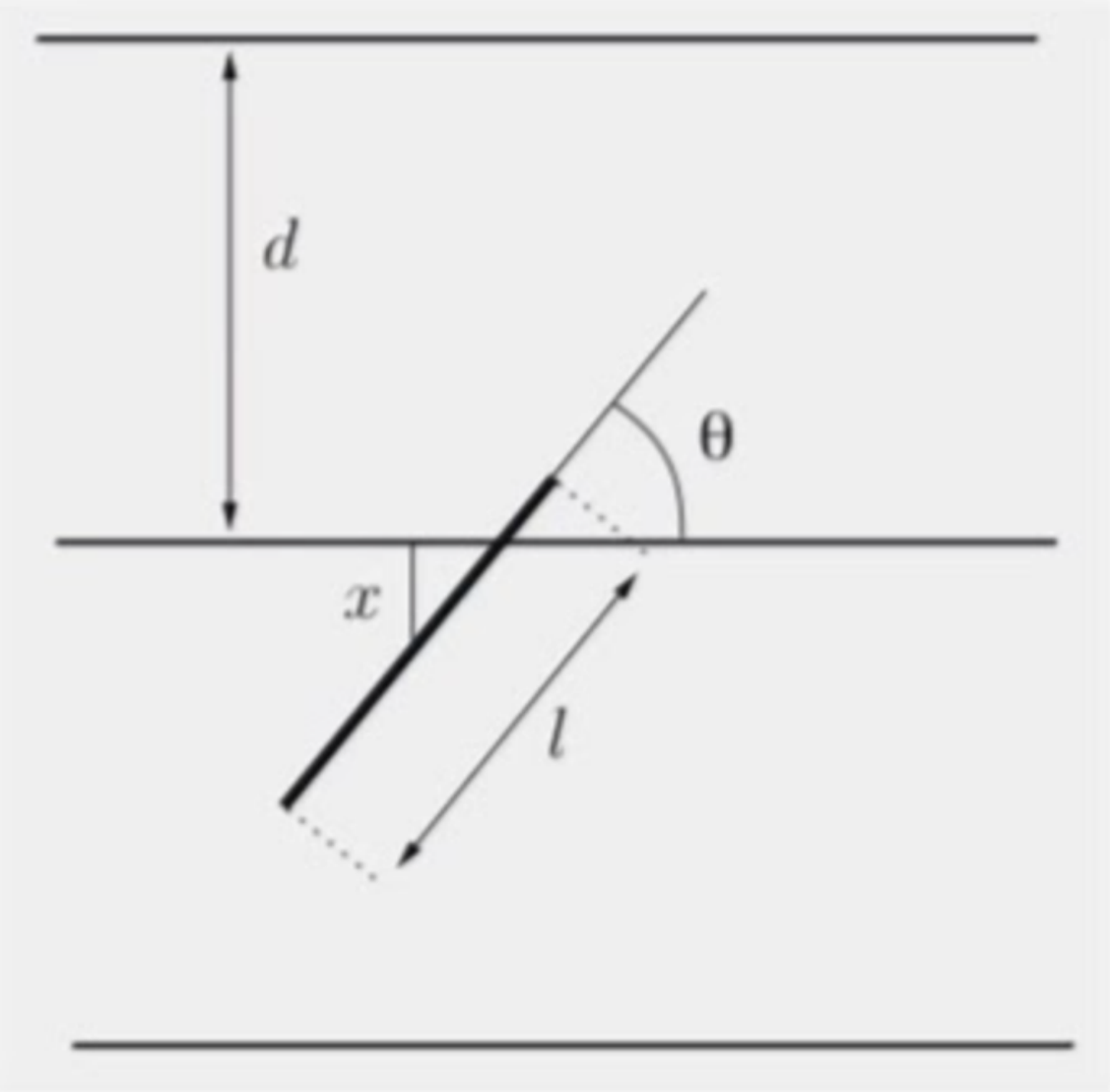


Figure: Geometry of Buffon's Needle

Buffon's Needle: Probability and Connection to π

Compute the crossing probability using the joint density of (X, Θ): $g(x, \theta) = \frac{2}{d}\frac{2}{\pi}$ for $0 \le x \le \frac{d}{2}$, $0 \le \theta \le \frac{\pi}{2}$

$$P(\text{cross}) = P\left(X \le \frac{\ell}{2}\sin\Theta\right) = \int_0^{\pi/2} \frac{2}{d} \cdot \frac{\ell}{2}\sin\theta \cdot \frac{2}{\pi}\, d\theta.$$

$$\Rightarrow \quad P(\text{cross}) = \frac{2\ell}{\pi d}.$$

Special Case: $\ell = d$

$$P(\text{cross}) = \frac{2}{\pi}.$$

Thus repeated trials can be used to **estimate** π:

$$\pi \approx \frac{2}{\hat{p}}, \qquad \hat{p} = \text{fraction of trials where needle crosses.}$$

Buffon's Needle = oldest Monte Carlo method for estimating π!